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广州市摩国技术服务有限公司
检测专用章

REPORT ON THE SURFACE SPREAD OF FLAME TEST TO BS 476 : PART 7 : 1997 ON "SEALING GASKET OF AIR DAMPER"

Client	:	Dongguan Tech Free Air Conditioning Mfy. Ltd.
Project	:	Fire Tests on Building Materials & Structure Part 7
Client Ref.	:	--
Report No.	:	G20849FU206211
Date of Report Issue	:	12 January 2021

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REPORT ON THE SURFACE SPREAD OF FLAME TEST

1.0 Introduction

Fugro Techinical Services (Guangzhou) Limited was commissioned by Dongguan Tech Free Air Conditioning Mfy. Ltd. to cunduct a test to determine the classification of the surface spread of flame of Sealing Gasket of Air Damper in accordance with BS 476-7: 1997 Incorporating Corrigendum No.1: 2014, "Fire tests on building materials and structures Part 7. Method of test to determine the classification of the surface spread of flame of products".

The test specimens were received by Fugro Technical Services (Guangzhou) Limited on 07 December 2020 and given Lab. Sample I.D.: FU206211.

The test was conducted on 17 December 2020 to 22 December 2020 by Fugro Technical Services (Guangzhou) Limited's approved sub-contract Laboratory.

2.0 Classification

In accordance with the class definitions given in BS 476-7: 1997 Incorporating Corrigendum No.1: 2014, the specimens tested are classified as **Class 2**.

The test results relate only to the behaviour of the test specimens of the product under the particular conditions of test; they are not intended to be the sole cirerion for assessing the protential fire hazard of the product in use.

The test results relate only to the specimens of the product in the form in which they were tested. Small differences in the composition or thickness of the product may significantlty affect the performance during the test and may therefore invalidate the test results. Care should be taken to ensure that any product which is supplied or used is fully represented by the specimens which were tested.

Check by: Eden

Approved Signatory:
Yan Cai Sheng
(Deputy General Manager)

Date of Report Issue : 12 January 2021

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3.0 Description of Test Specimens

Unless stated otherwise, all values are nominal, and all information in this Description of Test Specimens is supplied by the client.

Name and address of sponsor		Dongguan Tech Free Air Conditioning Mfy. Ltd. No. 11 Daji Industrial Zone, Shipai Town, Dongguan City, Guangdong Province, P.R.C.
Name and address of manufacturer/supplier		Dongguan Tech Free Air Conditioning Mfy. Ltd. No. 11 Daji Industrial Zone, Shipai Town, Dongguan City, Guangdong Province, P.R.C.
General description of overall product		Silicone Rubber
Product reference of overall product		Nil
Thickness of overall product		Approximately 15 mm
Density/Mass per unit area of overall product		N/A
Colour reference of overall product		Black/Silvery
Flame retardant details of overall product		Nil
Other details of overall product		Nil
Steel Frame	Generic type	Steel
	Product reference	Nil
	Manufacturer	Dongguan Tech Free Air Conditioning Mfy. Ltd.
	Colour	Silvery
	Thickness	Approximately 0.8 mm
	Density/Mass per unit area	Not provided
	Flame retardant details	N/A
	Other details	Nil
Sealing Gasket of Air Damper	Generic type	Sealing Gasket of Air Damper
	Product reference	Yi Zhong / SI-G092A/1mm / TF2011P9282
	Manufacturer	Dongguan Tech Free Air Conditioning Mfy. Ltd.
	Colour	Black
	Thickness	1 mm (wall-thickness)
	Density/Mass per unit area	N/A
	Flame retardant details	Nil
	Other details	Nil

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4.0 Test Details

Prior to the tests, all of the specimens were conditioned to constant mass at a temperature of 23±2℃ and a relative humidity of 50±5%, and maintained in this condition until requierd for testing.

The form of the tested specimens was assembly, as a fabrication of materials and/or composites that can contain air gaps.

The exposed surface of the tested specimens was the black surface.

5.0 Calibration and Test Procedures

The apparatus was calibrated in accordance with the requirement of clause 8 of the standard. The irradiance of the radiometer is as given in Table 1.

Table 1 - Irradiance along horizontal reference line on the calibration board

Distance along reference line from inside edge of specimen holder (mm)	Irradiance (kW/m^2)		
	specified	min.	max.
75	32.5	32.0	33.0
225	21.0	20.5	21.5
375	14.5	14.0	15.0
525	10.0	9.5	10.5
675	7.0	6.5	7.5
825	5.0	4.5	5.5

The test was conducted in accordance with the procedure specified in clause 9.1 of the standard.

Throughout the test, observations were made following clause 9.2 of the standard.

The behaviour of the product which results in the exposed surface not being available for the measurements of the spread of flame shall render the test on that specimen invalid, as explained in the clause 9.2.3 of the standard.

If one or more invalid test results are obtained, no more than nine specimens shall be tested in order to obtain the required six valid test results.

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6.0 Test Results

Specimen No.	FU206211/01	FU206211/02	FU206211/03	FU206211/04	FU206211/05	FU206211/06
Distance (mm)	Time of spread of flame to indicated distance (mm:ss)					
75	2:01	6:27	2:40	1:59	4:32	5:48
165	4:27	6:51	8:24	3:55	4:59	6:02
190	4:57	7:11	8:30	4:21	5:27	6:14
215	5:00	7:37	9:04	5:09	8:11	7:22
240	6:48	--	--	7:08	--	--
265	--	--	--	--	--	--
290	--	--	--	--	--	--
375	--	--	--	--	--	--
455	--	--	--	--	--	--
500	--	--	--	--	--	--
525	--	--	--	--	--	--
600	--	--	--	--	--	--
675	--	--	--	--	--	--
710	--	--	--	--	--	--
750	--	--	--	--	--	--
785	--	--	--	--	--	--
825	--	--	--	--	--	--
Maximum distance traveled at 1.5 min (mm)	25	0	0	25	0	0
Maximum distance traveled during the whole test (mm)	240	220	230	250	230	215
Time to reach maximum distance traveled (mm:ss)	6:48	7:40	10:00	7:30	10:00	7:22

Observations made during the test and comments on any difficulties encountered during the test	None.

Note: 1. "--" indicates that the flame did not reach the reference line.
2. Six specimens were tested and six valid test results were obtained.

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7.0 Classification Criteria

The results recorded in the test shall be used to obtain a classification for the product according to the criteria specified in table 2. At least five of the six test specimens for which valid test results have been obtained shall have a spread of flame which does not exceed the 1.5 min limit and the final limit specified for the designated class.

The remaining specimen shall not exceed this limit by more than the tolerances given in table 2.

Table 2 - Classification of spread of flame

Classification	Spread of flame at 1.5 min		Final spread of flame	
	Limit (mm)	Limit for one specimen in sample (mm)	Limit (mm)	Limit for one specimen in sample (mm)
Class 1	165	165 ± 25	165	165 ± 25
Class 2	215	215 ± 25	455	455 ± 25
Class 3	265	265 ± 25	710	710 ± 25
Class 4	Exceeding the limits for class 3			

Explanation of prefix and suffixes which may be added to the classification:

1. A suffix "R" is added to the classification if more than six specimens are required in order to obtain six valid test results (e.g. class 2R).

2. A prefix "D" is added to the classification of any product which does not comply with the surface characteristics specified in the standard and has therefore been tested in a modified form (e.g. class D3).

3. A suffix "Y" is added to the classification if any softening and/or other behaviour that may affect the flame spread occurs (e.g. class 3Y).

For example, a material could have a classification of D3RY indicating:

a) a modified surface has been used;

b) a class 3 result has been obtained;

c) additional specimens have been used to obtain six valid results; and

d) softening and/or other behaviour has occurred which is considered to have affected the test result.

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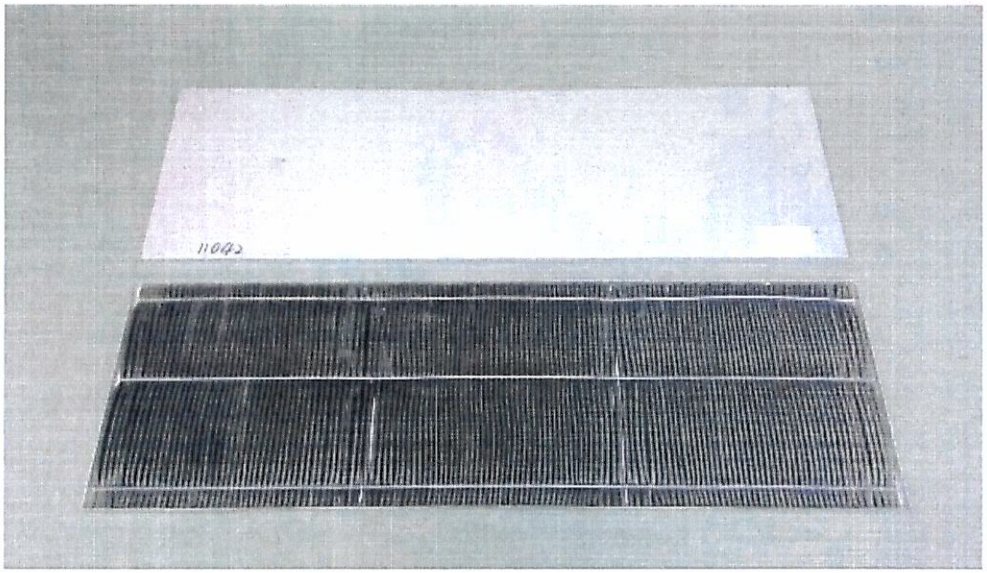
Appendix Photograph of Test Specimens



Sample I.D.: FU206211

** End of Report **

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